

JUNIOR MATHS

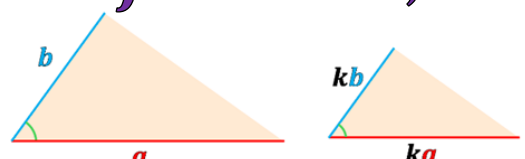
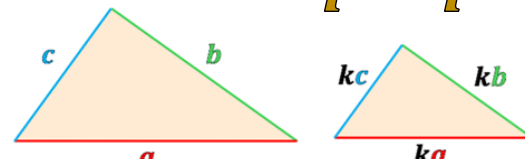
CH 17 (B): SIMILARITY

Key Learning Objectives

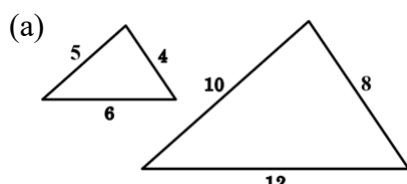
- ☐ Understand the concept of similarity. (Q1, 2)
- ☐ Apply properties of similarity (Triangles). (Q3-9, 13-18, 20, 21)
- ☐ Apply properties of similarity (2-D figures). (Q10-12, 19, 22, 23)



ratio of 2 sides, inc $\angle$ 3 sides prop.

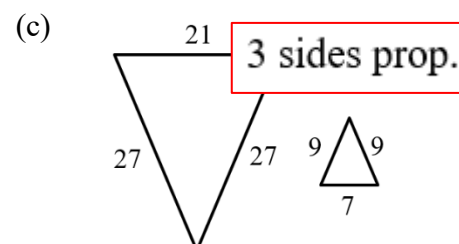
1. In each of the following, the two given triangles are similar. State the reasons for the similarity.



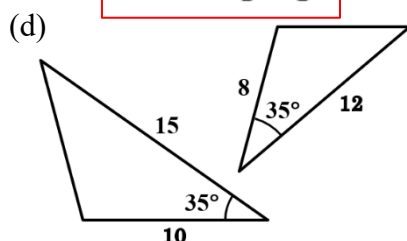
3 sides prop.



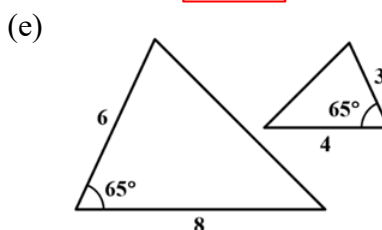
AAA



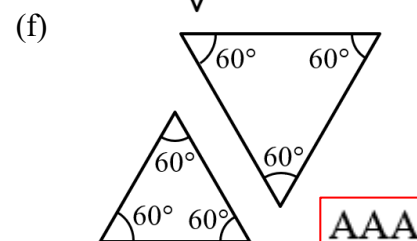
3 sides prop.



ratio of 2 sides, inc. $\angle$

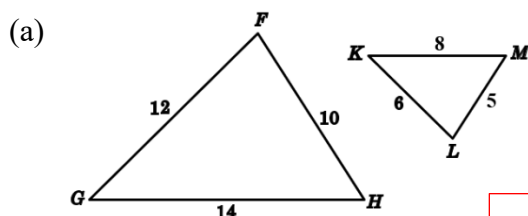


ratio of 2 sides, inc. $\angle$

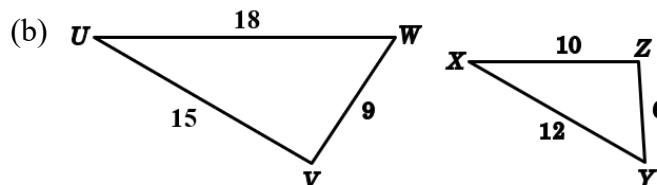


AAA

2. Determine whether each of the following pairs of triangles must be similar. If yes, give reasons.



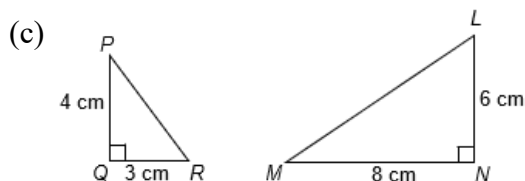
No.



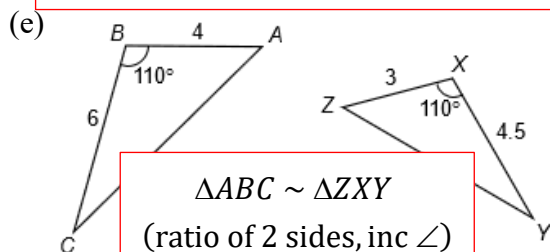
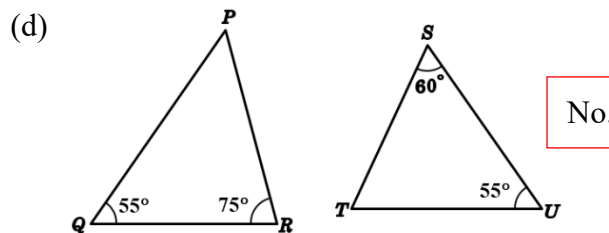
$\triangle UVW \sim \triangle XZY$ (3 sides prop.)

S2 JM17 – SIMILARITY (B)

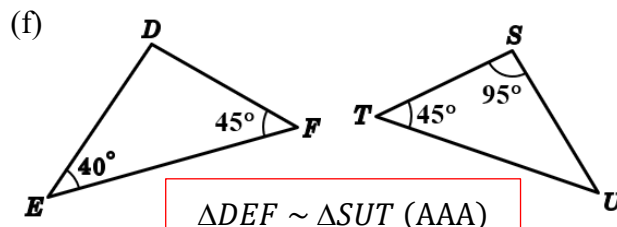
CONDITIONS FOR SIMILAR TRIANGLES



$\triangle PQR \sim \triangle MNL$ (ratio of 2 sides, inc $\angle$)

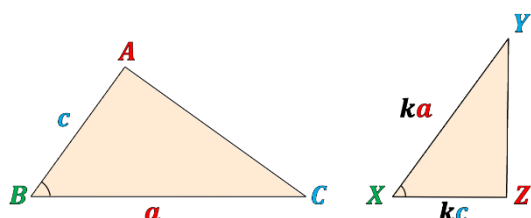


$\triangle ABC \sim \triangle ZXY$
(ratio of 2 sides, inc $\angle$)



$\triangle DEF \sim \triangle SUT$ (AAA)

Prove that $\triangle ABC \sim \triangle ZXY$



$$\frac{ZX}{AB} = \frac{kc}{c} = k$$

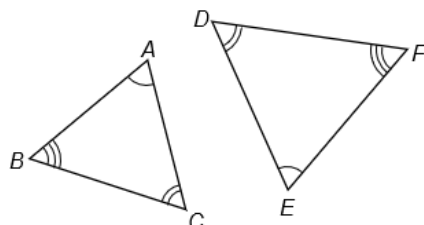
$$\frac{XY}{BY} = \frac{ka}{a} = k = \frac{ZX}{AB}$$

$$\angle B = \angle X \quad (\text{given})$$

$\therefore \triangle ABC \sim \triangle ZXY$ (ratio of 2 sides, inc $\angle$)

Refer to the given information in the figures to answer each of the following questions. (Q3-8)

3. Prove that $\triangle ABC \sim \triangle EFD$.



In $\triangle ABC$ and $\triangle EFD$,

$$\angle A = \angle E$$

given

$$\angle B = \angle F$$

given

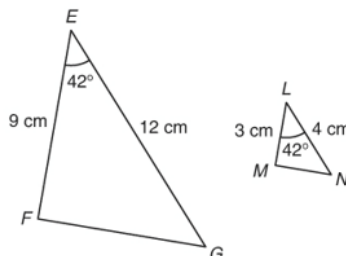
$$\angle C = \angle D$$

given

$\therefore \triangle ABC \sim \triangle EFD$

AAA

4. Prove that $\triangle EFG \sim \triangle LMN$.



In $\triangle EFG$ and $\triangle LMN$,

$$\therefore \frac{EF}{LM} = \frac{9 \text{ cm}}{3 \text{ cm}} = 3$$

$$\frac{EG}{LN} = \frac{12 \text{ cm}}{4 \text{ cm}} = 3$$

$$\therefore \frac{EF}{LM} = \frac{EG}{LN}$$

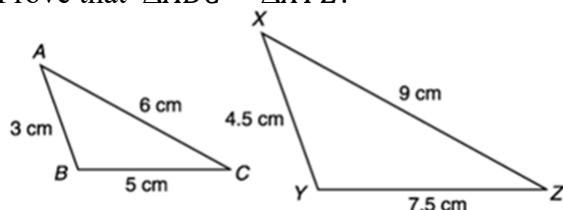
$$\angle E = \angle L$$

given

$\therefore \triangle EFG \sim \triangle LMN$

ratio of 2 sides, inc. $\angle$

5. Prove that $\triangle ABC \sim \triangle XYZ$.



In $\triangle ABC$ and $\triangle XYZ$,

$$\therefore \frac{AB}{XY} = \frac{3 \text{ cm}}{4.5 \text{ cm}} = \frac{2}{3}$$

$$\frac{AC}{XZ} = \frac{6 \text{ cm}}{9 \text{ cm}} = \frac{2}{3}$$

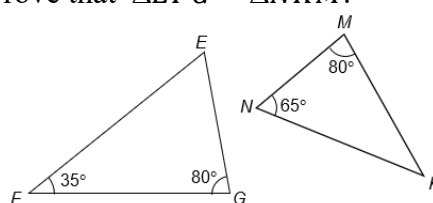
$$\frac{BC}{YZ} = \frac{5 \text{ cm}}{7.5 \text{ cm}} = \frac{2}{3}$$

$$\therefore \frac{AB}{XY} = \frac{AC}{XZ} = \frac{BC}{YZ}$$

$$\therefore \triangle ABC \sim \triangle XYZ$$

3 sides prop.

6. Prove that $\triangle EFG \sim \triangle NKM$.



In $\triangle EFG$ and $\triangle NKM$,

$$\angle G = \angle M$$

given

$$\therefore \angle E = 180^\circ - \angle F - \angle G$$

$$= 180^\circ - 35^\circ - 80^\circ$$

$$= 65^\circ$$

$$\therefore \angle E = \angle N$$

$\angle$ sum of $\triangle$

$$\therefore \angle K = 180^\circ - \angle M - \angle N$$

$$= 180^\circ - 80^\circ - 65^\circ$$

$$= 35^\circ$$

$$\therefore \angle F = \angle K$$

$\therefore \triangle EFG \sim \triangle NKM$

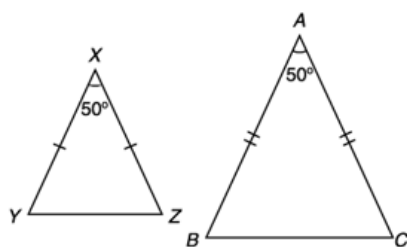
$\angle$ sum of $\triangle$

AAA

S2 JM17 – SIMILARITY (B)

CONDITIONS FOR SIMILAR TRIANGLES

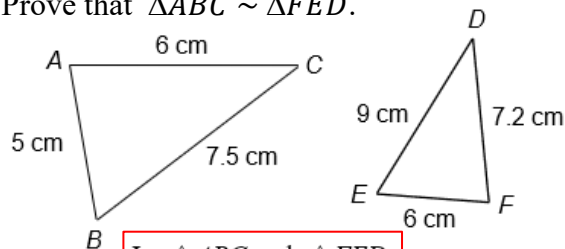
7. Prove that $\triangle XYZ \sim \triangle ABC$.



In $\triangle XYZ$ and $\triangle ABC$, $\therefore XY = XZ$ and $AB = AC$ $\therefore \frac{XY}{AB} = \frac{XZ}{AC}$ $\angle X = \angle A$ $\therefore \triangle XYZ \sim \triangle ABC$	given given given ratio of 2 sides, inc. $\angle$
---	--

Tip for Q9: Find the internal ratio for each triangle first. (e.g. long : short)

8. Prove that $\triangle ABC \sim \triangle FED$.



In $\triangle ABC$ and $\triangle FED$,

$$\therefore \frac{AB}{FE} = \frac{5 \text{ cm}}{9 \text{ cm}} = \frac{5}{9}$$

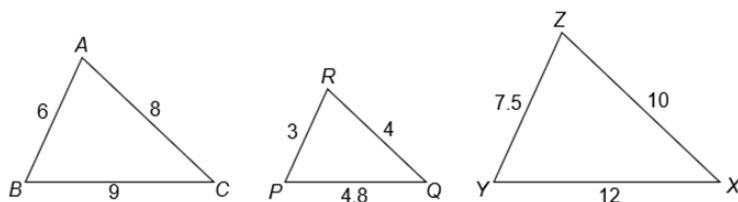
$$\frac{BC}{ED} = \frac{7.5 \text{ cm}}{6 \text{ cm}} = \frac{5}{4}$$

$$\frac{AC}{FD} = \frac{6 \text{ cm}}{7.2 \text{ cm}} = \frac{5}{6}$$

$\therefore \frac{AB}{FE} = \frac{BC}{ED} = \frac{AC}{FD}$	3 sides prop.
$\therefore \triangle ABC \sim \triangle FED$	

9. In each of the following figures, determine which two triangles must be similar and give reasons

(a)



In $\triangle PQR$ and $\triangle YXZ$,

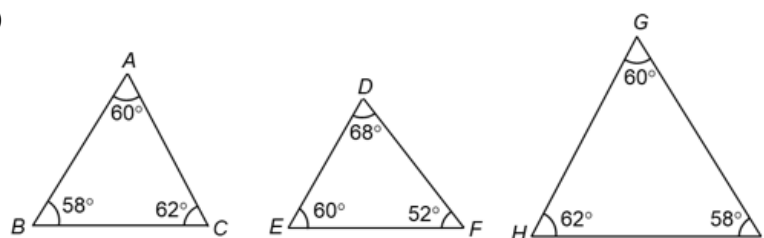
$$\therefore \frac{PQ}{YX} = \frac{4.8}{12} = \frac{2}{5}$$

$$\frac{QR}{XZ} = \frac{4}{10} = \frac{2}{5}$$

$$\frac{PR}{YZ} = \frac{3}{7.5} = \frac{2}{5}$$

$\therefore \frac{PQ}{YX} = \frac{QR}{XZ} = \frac{PR}{YZ}$	3 sides prop.
$\therefore \triangle PQR \sim \triangle YXZ$	

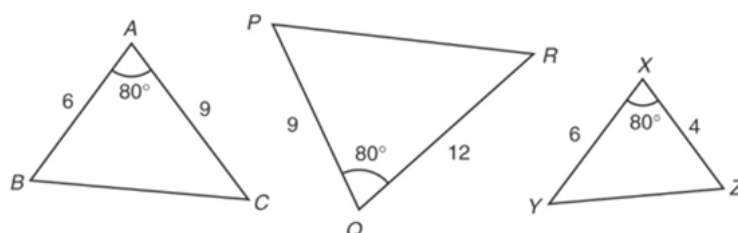
(b)



In $\triangle ABC$ and $\triangle GHI$,

$\angle A = \angle G$	given
$\angle B = \angle I$	given
$\angle C = \angle H$	given
$\therefore \triangle ABC \sim \triangle GHI$	AAA

(c)



$\angle A = \angle X$	given
$\therefore \triangle ABC \sim \triangle XYZ$	ratio of 2 sides, inc. $\angle$

In $\triangle ABC$ and $\triangle XYZ$,

$$\therefore \frac{AB}{XZ} = \frac{6}{4} = \frac{3}{2}$$

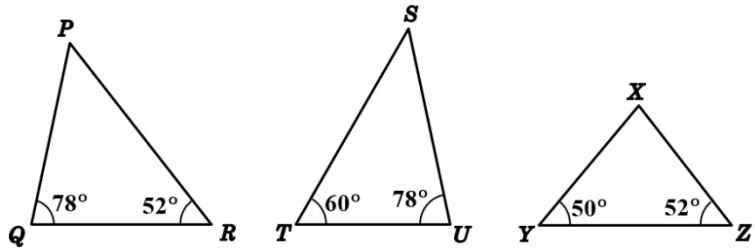
$$\frac{AC}{XY} = \frac{9}{6} = \frac{3}{2}$$

$$\therefore \frac{AB}{XZ} = \frac{AC}{XY}$$

S2 JM17 – SIMILARITY (B)

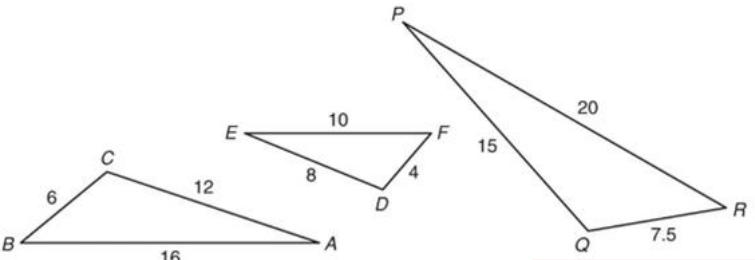
CONDITIONS FOR SIMILAR TRIANGLES

(d)



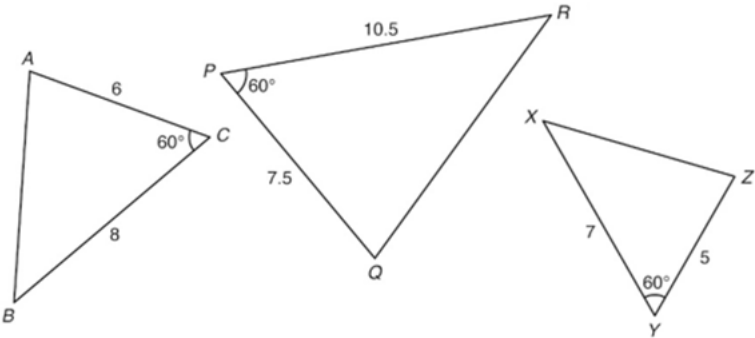
In $\triangle PQR$ and $\triangle YXZ$, $\angle R = \angle Z = 52^\circ$ In $\triangle PQR$ $\angle P = 180^\circ - 78^\circ - 52^\circ$ $= 50^\circ$ $= \angle Y$ $\therefore \triangle PQR \sim \triangle YXZ$	given $\angle$ sum of Δ AA
---	--

(e)



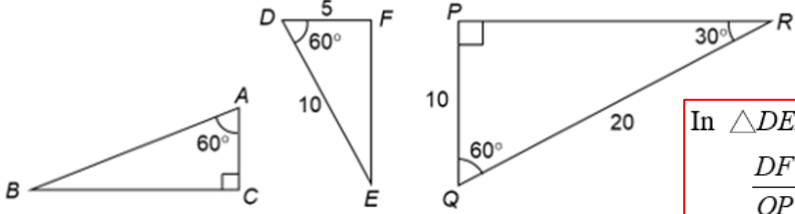
In $\triangle ACB$ and $\triangle PQR$, $\therefore \frac{AC}{PQ} = \frac{12}{15} = \frac{4}{5}$ $\frac{CB}{QR} = \frac{6}{7.5} = \frac{4}{5}$ $\frac{AB}{PR} = \frac{16}{20} = \frac{4}{5}$	$\therefore \frac{AC}{PQ} = \frac{CB}{QR} = \frac{AB}{PR}$ $\therefore \triangle ACB \sim \triangle PQR$ 3 sides prop.
---	--

(f)



In $\triangle PQR$ and $\triangle YZX$, $\therefore \frac{PQ}{YZ} = \frac{7.5}{5} = \frac{3}{2}$ $\frac{PR}{YX} = \frac{10.5}{7} = \frac{3}{2}$ $\therefore \frac{PQ}{YZ} = \frac{PR}{YX}$	$\angle P = \angle Y$ $\therefore \triangle PQR \sim \triangle YZX$ given ratio of 2 sides, inc. $\angle$
---	---

◆ 10. Prove that the three triangles in the figure are similar.



In $\triangle ABC$ and $\triangle QRP$, $\angle A = \angle Q$ given $\angle C = \angle P$ given $\therefore \angle A + \angle B + \angle C = 180^\circ$ $\angle$ sum of Δ $60^\circ + \angle B + 90^\circ = 180^\circ$ $\angle B = 30^\circ$ $\therefore \angle B = \angle R$ $\therefore \triangle ABC \sim \triangle QRP$ AAA	In $\triangle DEF$ and $\triangle QRP$, $\frac{DF}{QP} = \frac{5}{10} = \frac{1}{2}$ $\frac{DE}{QR} = \frac{10}{20} = \frac{1}{2}$ $\angle D = \angle Q$ given $\therefore \triangle DEF \sim \triangle QRP$ ratio of 2 sides, inc. $\angle$
---	--

$\therefore \triangle ABC, \triangle DEF$ and $\triangle QRP$ are similar.

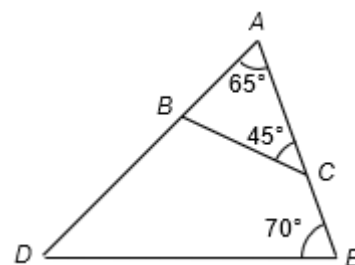
S2 JM17 – SIMILARITY (B)

CONDITIONS FOR SIMILAR TRIANGLES

Level 2

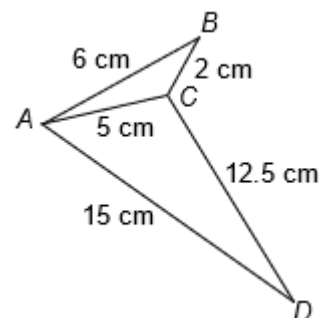
11. In the figure, ABD and ACE are straight lines. Prove that $\triangle ABC \sim \triangle AED$.

In $\triangle ABC$ and $\triangle AED$, $\angle ABC + \angle ACB + \angle BAC = 180^\circ$ $\angle ABC + 45^\circ + 65^\circ = 180^\circ$ $\angle ABC = 70^\circ$ $\therefore \angle ABC = \angle AED$	$\angle$ sum of $\triangle$
$\angle ADE + \angle AED + \angle DAE = 180^\circ$ $\angle ADE + 70^\circ + 65^\circ = 180^\circ$ $\angle ADE = 45^\circ$ $\therefore \angle ACB = \angle ADE$ $\angle BAC = \angle EAD$ $\therefore \triangle ABC \sim \triangle AED$	$\angle$ sum of $\triangle$ common $\angle$ AAA



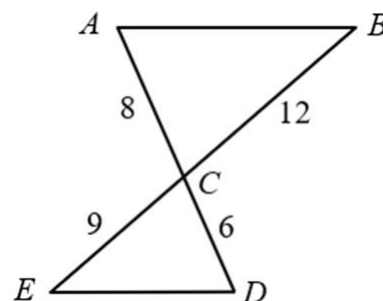
12. Refer to the figure. Prove that $\triangle ABC \sim \triangle DAC$.

In $\triangle ABC$ and $\triangle DAC$, $\therefore \frac{AB}{DA} = \frac{6 \text{ cm}}{15 \text{ cm}} = \frac{2}{5}$ $\frac{BC}{AC} = \frac{2 \text{ cm}}{5 \text{ cm}} = \frac{2}{5}$ $\frac{AC}{DC} = \frac{5 \text{ cm}}{12.5 \text{ cm}} = \frac{2}{5}$ $\therefore \frac{AB}{DA} = \frac{BC}{AC} = \frac{AC}{DC}$ $\therefore \triangle ABC \sim \triangle DAC$	3 sides prop.
---	---------------



13. In the figure, ACD and BCE are straight lines. Prove that $\triangle ABC \sim \triangle DEC$.

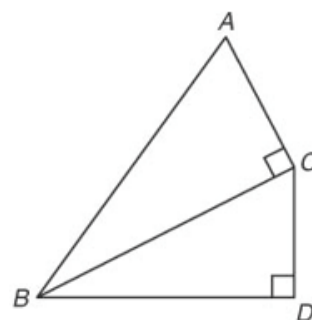
In $\triangle ABC$ and $\triangle DEC$, $\angle ACB = \angle DCE$ $\frac{AC}{DC} = \frac{8}{6} = \frac{4}{3}$ $\frac{BC}{EC} = \frac{12}{9} = \frac{4}{3} = \frac{AC}{DC}$ $\therefore \triangle ABC \sim \triangle DEC$	vert. opp. $\angle$ s $\angle$ sum of $\triangle$ ratio of 2 sides, inc. $\angle$
---	---



- ◆ 14. In the figure, BC bisects $\angle ABD$ and $\angle ACB = \angle CDB = 90^\circ$.

- (a) Prove that $\triangle ABC \sim \triangle CBD$.
 (b) If $BD = 8 \text{ cm}$ and $BC = 10 \text{ cm}$, find BA .

(a) In $\triangle ABC$ and $\triangle CBD$, $\therefore BC$ bisects $\angle ABD$. $\therefore \angle ABC = \angle CBD$ $\angle ACB = \angle CDB = 90^\circ$ $\therefore \angle BAC = 180^\circ - \angle ACB - \angle ABC$ $\angle CBD = 180^\circ - \angle CDB - \angle CBD$ $\therefore \angle BAC = \angle CBD$ $\therefore \triangle ABC \sim \triangle CBD$	given $\angle$ sum of $\triangle$ $\angle$ sum of $\triangle$ AAA
(b) $\therefore \triangle ABC \sim \triangle CBD$ (proved in (a)) $\therefore \frac{BA}{BC} = \frac{BC}{BD}$ (corr. sides, $\sim \triangle$ s) $\frac{BA}{10 \text{ cm}} = \frac{10 \text{ cm}}{8 \text{ cm}}$ $BA = \underline{\underline{10.5 \text{ cm}}}$	



S2 JM17 – SIMILARITY (B)

CONDITIONS FOR SIMILAR TRIANGLES

15. In the figure, BCD is a straight line. Prove that $\triangle ABC \sim \triangle DBA$.

In $\triangle ABC$ and $\triangle DBA$,

$$\therefore \frac{AB}{DB} = \frac{4 \text{ cm}}{(2+6) \text{ cm}} = \frac{1}{2}$$

$$\frac{BC}{BA} = \frac{2 \text{ cm}}{4 \text{ cm}} = \frac{1}{2}$$

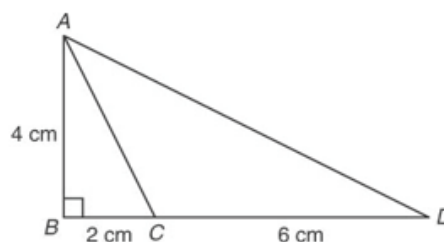
$$\therefore \frac{AB}{DB} = \frac{BC}{BA}$$

$$\angle ABC = \angle DBA$$

$$\therefore \triangle ABC \sim \triangle DBA$$

common $\angle$

ratio of 2 sides, inc. $\angle$



16. In the figure, ADC is a straight line and $\angle ABD = \angle ACB$.

(a) Prove that $\triangle ABD \sim \triangle ACB$.

(b) Find DC .

(a) In $\triangle ABD$ and $\triangle ACB$,

$$\angle BAD = \angle CAB$$

$$\angle ABD = \angle ACB$$

$$\therefore \angle BDA = 180^\circ - \angle BAD - \angle ABD$$

$$\angle CBA = 180^\circ - \angle CAB - \angle ACB$$

$$\therefore \angle BDA = \angle CBA$$

$$\therefore \triangle ABD \sim \triangle ACB$$

common $\angle$
given

$\angle$ sum of $\triangle$

$\angle$ sum of $\triangle$

AAA

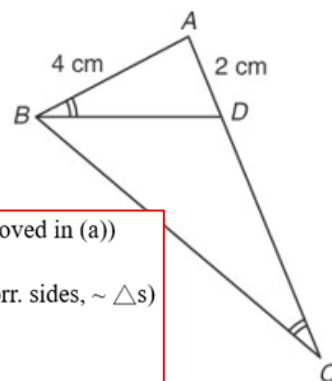
(b) $\therefore \triangle ABD \sim \triangle ACB$ (proved in (a))

$$\therefore \frac{AC}{AB} = \frac{AB}{AD} \quad (\text{corr. sides, } \sim \triangle s)$$

$$\frac{DC + 2 \text{ cm}}{4 \text{ cm}} = \frac{4 \text{ cm}}{2 \text{ cm}}$$

$$DC + 2 \text{ cm} = 8 \text{ cm}$$

$$DC = \underline{6 \text{ cm}}$$



17. In the figure, D and E are points lying on AC and BC respectively. Prove that $\triangle ABC \sim \triangle DEB$.

In $\triangle ABC$,

$$\angle ABC + \angle BAC + \angle BCA = 180^\circ \quad (\angle \text{ sum of } \triangle)$$

$$(\angle DBE + 62^\circ) + 26^\circ + 46^\circ = 180^\circ$$

$$\angle DBE = 46^\circ$$

In $\triangle BED$,

$$\angle BED + \angle DBE + \angle EDB = 180^\circ \quad (\angle \text{ sum of } \triangle)$$

$$\angle BED + 46^\circ + 26^\circ = 180^\circ$$

$$\angle BED = 108^\circ$$

In $\triangle ABC$ and $\triangle DEB$,

$$\therefore \angle ABC = 62^\circ + 46^\circ$$

$$= 108^\circ$$

$$\therefore \angle ABC = \angle DEB$$

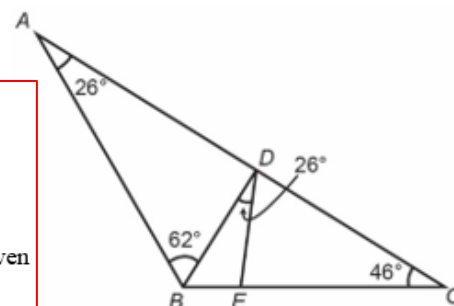
$$\angle BAC = \angle EDB$$

$$\angle ACB = \angle DBE$$

$$\therefore \triangle ABC \sim \triangle DEB$$

given

AAA



18. In the figure, $AECF$ is a straight line and $\angle ABC = \angle FDE$.

(a) Prove that $\triangle ABC \sim \triangle FDE$.

(b) Find EC .

(a) In $\triangle ABC$ and $\triangle FDE$,

$$\therefore \frac{AB}{FD} = \frac{20 \text{ cm}}{15 \text{ cm}} = \frac{4}{3}$$

$$\frac{BC}{DE} = \frac{8 \text{ cm}}{6 \text{ cm}} = \frac{4}{3}$$

$$\therefore \frac{AB}{FD} = \frac{BC}{DE}$$

$$\angle ABC = \angle FDE$$

$$\therefore \triangle ABC \sim \triangle FDE$$

given

ratio of 2 sides, inc. $\angle$

(b) $\therefore \triangle ABC \sim \triangle FDE$ (proved in (a))

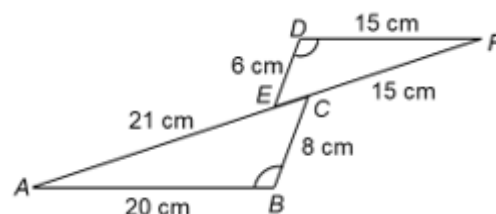
$$\therefore \frac{AC}{FE} = \frac{BC}{DE} \quad (\text{corr. sides, } \sim \triangle s)$$

$$\frac{EC + 21 \text{ cm}}{EC + 15 \text{ cm}} = \frac{4}{3}$$

$$3(EC + 21 \text{ cm}) = 4(EC + 15 \text{ cm})$$

$$3EC + 63 \text{ cm} = 4EC + 60 \text{ cm}$$

$$EC = \underline{3 \text{ cm}}$$



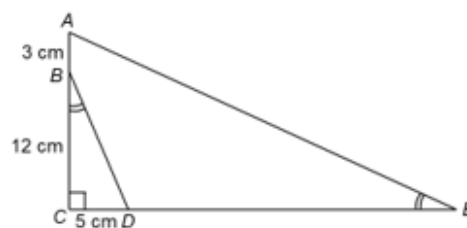
S2 JM17 – SIMILARITY (B)

CONDITIONS FOR SIMILAR TRIANGLES

19. In the figure, ABC and CDE are straight lines. $AC \perp CE$ and $\angle AEC = \angle DBC$.

(a) Prove that $\triangle ACE \sim \triangle DCB$.

(b) Find the area of quadrilateral $ABDE$.



(a) In $\triangle ACE$ and $\triangle DCB$,	
$\angle ACE = \angle DCB$	common $\angle$
$\angle AEC = \angle DBC$	given
$\therefore \angle EAC = 180^\circ - \angle ACE - \angle AEC$	$\angle$ sum of $\triangle$
$\angle BDC = 180^\circ - \angle DCB - \angle DBC$	$\angle$ sum of $\triangle$
$\therefore \angle EAC = \angle BDC$	
$\therefore \triangle ACE \sim \triangle DCB$	AAA

(b) $\therefore \triangle ACE \sim \triangle DCB$	(proved in (a))
$\therefore \frac{CE}{CB} = \frac{AC}{DC}$	(corr. sides, $\sim \triangle$ s)
$\frac{5 \text{ cm} + DE}{12 \text{ cm}} = \frac{(3+12) \text{ cm}}{5 \text{ cm}}$	
$\frac{5 \text{ cm} + DE}{12 \text{ cm}} = 3$	
$5 \text{ cm} + DE = 36 \text{ cm}$	
$DE = 31 \text{ cm}$	

$$\begin{aligned}
 \text{Area of quadrilateral } ABDE &= \text{area of } \triangle ACE - \text{area of } \triangle DCB \\
 &= \frac{1}{2} \times CE \times AC - \frac{1}{2} \times CD \times BC \\
 &= \left[\frac{1}{2} \times (5+31) \times (3+12) - \frac{1}{2} \times 5 \times 12 \right] \text{ cm}^2 \\
 &= \underline{\underline{240 \text{ cm}^2}}
 \end{aligned}$$

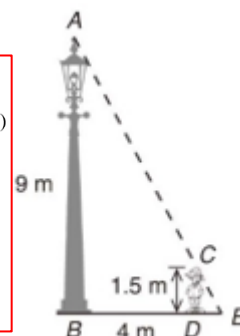
20. In the figure, AB is a lamppost of height 9 m. A boy CD of height 1.5 m is standing 4 m away from the lamppost. It is given that ACE and BDE are straight lines, and $\angle ABE = \angle CDE = 90^\circ$.

(a) Prove that $\triangle ABE \sim \triangle CDE$.

(b) Find the length of shadow DE .

(a) In $\triangle ABE$ and $\triangle CDE$,	
$\angle ABE = \angle CDE$	given
$\angle AEB = \angle CED$	common $\angle$
$\therefore \angle BAE = 180^\circ - \angle ABE - \angle AEB$	$\angle$ sum of $\triangle$
$\angle DCE = 180^\circ - \angle CDE - \angle CED$	$\angle$ sum of $\triangle$
$\therefore \angle BAE = \angle DCE$	
$\therefore \triangle ABE \sim \triangle CDE$	AAA

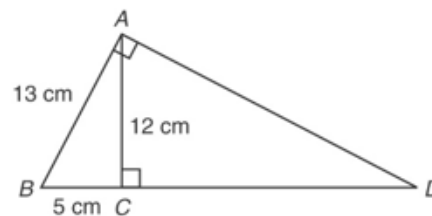
(b) $\therefore \triangle ABE \sim \triangle CDE$	(proved in (a))
$\therefore \frac{DE}{BE} = \frac{CD}{AB}$	(corr. sides, $\sim \triangle$ s)
$\frac{DE}{DE + 4 \text{ m}} = \frac{1.5 \text{ m}}{9 \text{ m}}$	
$9DE = 1.5DE + 6 \text{ m}$	
$7.5DE = 6 \text{ m}$	
$DE = 0.8 \text{ m}$	
$\therefore$ The length of shadow DE is 0.8 m.	



21. In the figure, $\angle BAD = \angle ACD = 90^\circ$, $AB = 13 \text{ cm}$, $BC = 5 \text{ cm}$ and $AC = 12 \text{ cm}$.

(a) Prove that $\triangle ABC \sim \triangle DBA \sim \triangle DAC$.

(b) Find DA and CD .



(a) In $\triangle ABC$ and $\triangle DBA$,	
$\angle ACB = \angle DAB$	given
$\angle ABC = \angle DBA$	common $\angle$
$\therefore \angle BAC = 180^\circ - \angle ACB - \angle ABC$	$\angle$ sum of $\triangle$
$\angle BDA = 180^\circ - \angle DAB - \angle DBA$	$\angle$ sum of $\triangle$
$\therefore \angle BAC = \angle BDA$	
$\therefore \triangle ABC \sim \triangle DBA$	AAA
In $\triangle DAC$ and $\triangle DBA$,	
$\angle ACD = \angle BAD$	given
$\angle ADC = \angle BDA$	common $\angle$
$\therefore \angle CAD = 180^\circ - \angle ACD - \angle ADC$	$\angle$ sum of $\triangle$
$\angle ABD = 180^\circ - \angle BAD - \angle BDA$	$\angle$ sum of $\triangle$
$\therefore \angle CAD = \angle ABD$	
$\therefore \triangle DAC \sim \triangle DBA$	AAA
$\therefore \triangle ABC \sim \triangle DBA \sim \triangle DAC$	

(b) $\therefore \triangle DAC \sim \triangle ABC$	(proved in (a))
$\therefore \frac{DA}{AB} = \frac{AC}{BC}$	(corr. sides, $\sim \triangle$ s)
$\frac{DA}{13 \text{ cm}} = \frac{12 \text{ cm}}{5 \text{ cm}}$	
$DA = 31.2 \text{ cm}$	
$\frac{CD}{CA} = \frac{CA}{CB}$	(corr. sides, $\sim \triangle$ s)
$\frac{CD}{12 \text{ cm}} = \frac{12 \text{ cm}}{5 \text{ cm}}$	
$CD = 28.8 \text{ cm}$	

S2 JM17 – SIMILARITY (B)

CONDITIONS FOR SIMILAR TRIANGLES

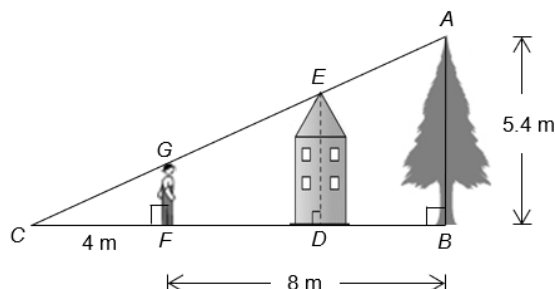
22. In the figure, AB is a tree of height 5.4 m. A man is standing 8 m away from the tree. There is a house between the man and the tree. It is given that $AEGC$ and $BDFC$ are straight lines and $CF = 4$ m.

(a) Prove that $\triangle ABC \sim \triangle EDC \sim \triangle GFC$.

(b) Find the height of the man.

(c) If the height of the house is 0.225 m more than two times the height of the man, find the distance between the house and the tree.

(a) In $\triangle ABC$ and $\triangle EDC$, $\angle ACB = \angle ECD$ $\angle ABC = \angle EDC$ $\therefore \angle BAC = 180^\circ - \angle ACB - \angle ABC$ $\angle CED = 180^\circ - \angle ECD - \angle EDC$ $\therefore \angle BAC = \angle CED$ $\therefore \triangle ABC \sim \triangle EDC$ In $\triangle ABC$ and $\triangle GFC$, $\angle ACB = \angle GCF$ $\angle ABC = \angle GFC$ $\therefore \angle BAC = 180^\circ - \angle ACB - \angle ABC$ $\angle FGC = 180^\circ - \angle GCF - \angle GFC$ $\therefore \angle BAC = \angle FGC$ $\therefore \triangle ABC \sim \triangle GFC$ $\therefore \triangle ABC \sim \triangle EDC \sim \triangle GFC$	common $\angle$ given $\angle$ sum of $\triangle$ $\angle$ sum of $\triangle$ AAA common $\angle$ given $\angle$ sum of $\triangle$ $\angle$ sum of $\triangle$ AAA
--	---



(b) $\therefore \triangle ABC \sim \triangle GFC$ (proved in (a))

$$\therefore \frac{GF}{AB} = \frac{CF}{CB}$$

(corr. sides, $\sim \triangle$ s)

$$\frac{GF}{5.4 \text{ m}} = \frac{4 \text{ m}}{(4+8) \text{ m}}$$

$$GF = 1.8 \text{ m}$$

$\therefore$ The height of the man is 1.8 m.

(c) $ED = (2 \times 1.8 + 0.225) \text{ m} = 3.825 \text{ m}$

$$\therefore \triangle ABC \sim \triangle EDC$$
 (proved in (a))
$$\therefore \frac{CD}{CB} = \frac{ED}{AB}$$
 (corr. sides, $\sim \triangle$ s)
$$\frac{CD}{(4+8) \text{ m}} = \frac{3.825 \text{ m}}{5.4 \text{ m}}$$

$$\frac{CD}{12 \text{ m}} = \frac{17 \text{ m}}{24 \text{ m}}$$

$$CD = 8.5 \text{ m}$$

$$DB = CB - CD$$

$$= (12 - 8.5) \text{ m}$$

$$= 3.5 \text{ m}$$

$\therefore$ The distance between the house and the tree is 3.5 m.